Farzaneh Barat

Ph.D. Student in Mechanical Engineering

Lithium-Ion Batteries | **Battery Modeling** | **State Estimation** | **Machine Learning** | **Controls**

Lawrence, KS barat@ku.edu 785-424-2521

farzanehbarat.github.io Google Scholar LinkedIn GitHub

Summary

Ph.D. student developing physics-based and machine-learning methods for lithium-ion battery state estimation, parameter identification, and thermal management, with hands-on battery testing experience (HPPC, cycling, second-life modules) and 5 years of prior industrial R&D.

Education

University of Kansas **Lawrence, KS**
Ph.D. in Mechanical Engineering *Aug. 2023 – Present*

- **Advisors:** Prof. Huazhen Fang and Prof. Rachel Carter
- **Research Areas:** Lithium-ion battery modeling, estimation, parameter identification, electro-thermal systems, thermal management, machine learning, and second-life batteries

University of Kashan **Kashan, Iran**
M.Sc. in Mechanical Engineering *Sep. 2015 – Feb. 2018*

- Research in nonlinear optimal control and path tracking of robotic manipulators.

University of Kashan **Kashan, Iran**
B.Sc. in Mechanical Engineering *Sep. 2011 – Sep. 2015*

Battery, Controls & Machine-Learning Research

University of Kansas **Lawrence, KS**
Graduate Research Assistant *Aug. 2023 – Present*

- Develop physics-based and machine-learning methods for lithium-ion battery state estimation, parameter identification, thermal management, and health diagnostics (ACC 2026, CDC 2026).
- Design and run battery experiments (dynamic charge/discharge profiles, capacity and HPPC tests, thermal measurements) on fresh and second-life modules using PEC battery cyclers.

Uncertainty-Aware Parameter Identification of Battery Models *2024 – 2026*

- Formulated battery model calibration as a Bayesian inverse problem solved with gradient-free **Ensemble Kalman Inversion**, estimating model parameters together with their uncertainty.
- Identified equivalent-circuit and electro-thermal model parameters from voltage and temperature data; generalized the method to nonlinear state-space models (ACC 2026; IEEE TCST under review).

Recursive Parameter Identification for Nonlinear Battery Models *2025 – 2026*

- Developed a sequentialized Ensemble Kalman Inversion method that couples ensemble parameter updates with particle-based state inference for online identification of nonlinear stochastic state-space models with unmeasured states (CDC 2026).
- Applied the method to lithium-ion battery equivalent-circuit models using simulated and experimental data, benchmarking against nonlinear filtering and dual-estimation methods.

Neural-ODE-Enhanced Observer for Robust Battery SOC Estimation *2026*

- Developed a hybrid physics–machine-learning observer that augments a nonlinear electro-thermal battery observer with a **neural ODE** learning the evolution of SOC-estimation error under model-parameter uncertainty (submitted to ACC 2027).

Dynamic Thermal Management and Smart Charging of Lithium-Ion Batteries *2026 – Present*

- Develop model-based cooling and smart-charging strategies on coupled electro-thermal models to reduce temperature nonuniformity and degradation in cylindrical and pouch cells.

Second-Life Lithium-Ion Battery Characterization and Modeling *2026 – Present*

- Characterize retired modules (capacity, dynamic cycling, HPPC, thermal) to evaluate degradation, safety, and reuse potential, supporting physics-informed modeling, diagnostics, and RUL assessment.

Probabilistic Solver for PMP-Based Optimal Control *2025 – Present*

- Developing an uncertainty-aware solver that casts **Pontryagin's Minimum Principle** optimal control problems as boundary-value problems solved with probabilistic estimation and smoothing.

Technical Skills

Battery Modeling & BMS: Lithium-ion battery modeling, equivalent-circuit models, electro-thermal modeling, state-of-charge (SOC) estimation, parameter identification, degradation analysis, second-life battery systems, battery diagnostics, smart charging, thermal management

State Estimation & Controls: Nonlinear observers, Kalman filtering, Ensemble Kalman methods, particle filtering, state-space modeling, system identification, optimal control, numerical optimization

Programming & Machine Learning: Python, PyTorch, NumPy, SciPy, Pandas, MATLAB, Simulink, neural networks, neural ODEs, scientific computing, numerical integration, Monte Carlo simulation

Battery Testing & Experimentation: Battery cycling, charge/discharge testing, capacity characterization, HPPC testing, SOC characterization, thermal measurements, degradation testing, second-life battery-module evaluation, PEC battery cyclers

Engineering Software: ANSYS, Abaqus, SolidWorks, Autodesk Inventor, AutoCAD, CATIA, LaTeX

Publications

Journal Articles

- **F. Barat** and H. Fang, "System Identification of Nonlinear State-Space Models Using Ensemble Kalman Inversion with Application to Lithium-Ion Batteries," under review at *IEEE Transactions on Control Systems Technology*. [arXiv]
- M. Irani Rahaghi and **F. Barat**, "Solving Nonlinear Optimal Path Tracking Problem Using a New Closed-Loop Direct–Indirect Optimization Method: Application on Mechanical Manipulators," *Robotica*, vol. 37, no. 1, pp. 39–61, 2019. [DOI]

Conference Papers

- **F. Barat**, R. Carter, and H. Fang, "Robust State-of-Charge Estimation for Lithium-Ion Batteries by Neural Network-Enhanced Observer Design," submitted to *American Control Conference (ACC)*, 2027.
- **F. Barat**, R. Carter, S. Wilson, and H. Fang, "Recursive Parameter Identification of Nonlinear Stochastic State-Space Models via Sequentialized Ensemble Kalman Inversion," accepted to *IEEE Conference on Decision and Control (CDC)*, 2026. [arXiv]
- **F. Barat**, S. Wilson, H. Kim, and H. Fang, "System Identification of Lithium-Ion Battery Equivalent Circuit Models Using Ensemble Kalman Inversion," *American Control Conference (ACC)*, 2026. [IEEE Xplore] [arXiv]

Industry Experience

Diana Tajhiz Spadana Company / SEMAG Company **Isfahan, Iran**
Head of R&D / Mechanical Designer *Apr. 2018 – Jun. 2023*

- Led R&D and mechanical design of industrial products from concept through prototyping and production, using SolidWorks, Inventor, AutoCAD, and CATIA.

Teaching Experience

University of Kansas **Lawrence, KS**
Graduate Teaching Assistant *Spring 2025*

- Supported Numerical Analysis and Life Cycle Assessment labs (MATLAB, numerical methods).

University of Kashan **Kashan, Iran**
Graduate Teaching Assistant / Instructor *2015 – 2018*

Honors & Awards

- **3rd Place**, University of Kansas Graduate Student Research Showcase, 2025
- Ranked **4th** in M.Sc. Mechanical Engineering cohort, University of Kashan
- Ranked **1st** in B.Sc. Heat and Fluid Mechanics concentration, University of Kashan